

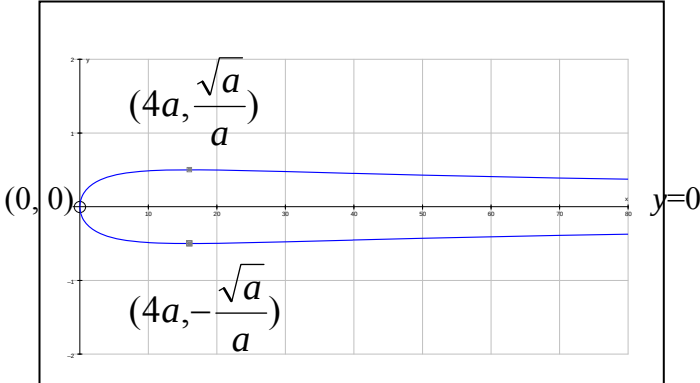
JC 2 H2 Mathematics Preliminary Examination 2009
Paper 1 -- Marking Scheme

1	$x^2 - 3x - 4 < 0$ $(x+1)(x-4) < 0$ $-1 < x < 4$ $\int_{-1}^{3\sqrt{2}} x^2 - 3x - 4 dx = \left \int_{-1}^4 x^2 - 3x - 4 dx \right + \int_4^{3\sqrt{2}} x^2 - 3x - 4 dx$ [Mark awarded for splitting correctly into 2 sections either using modulus or negative sign] $\int_{-1}^{3\sqrt{2}} x^2 - 3x - 4 dx = \left[\frac{x^3}{3} - \frac{3x^2}{2} - 4x \right]_{-1}^4 + \left[\frac{x^3}{3} - \frac{3x^2}{2} - 4x \right]_4^{3\sqrt{2}}$ [Mark awarded for correct integration] $\int_{-1}^{3\sqrt{2}} x^2 - 3x - 4 dx = 12\frac{1}{2} + 6\sqrt{2}$ OR $\int_{-1}^{3\sqrt{2}} x^2 - 3x - 4 dx = \frac{25 + 12\sqrt{2}}{2}$	[B1] [M1] [M1] [A1]
2(i)	$\overrightarrow{OM} = \frac{2\overrightarrow{OB} + \overrightarrow{OA}}{3}$ $= \frac{2 \begin{pmatrix} 3 \\ 3 \\ -3 \end{pmatrix} + \begin{pmatrix} -3 \\ 0 \\ 9 \end{pmatrix}}{3}$ $= \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}$	[M1] [A1]
(ii)	If OM is perpendicular to AB, $\overrightarrow{OM} \cdot \overrightarrow{AB} = 0$ $\overrightarrow{AB} = \begin{pmatrix} 6 \\ 3 \\ -12 \end{pmatrix}$ $\overrightarrow{OM} \cdot \overrightarrow{AB} = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 6 \\ 3 \\ -12 \end{pmatrix} = 0$	[M1] [M1]

	LHS $= \sum_{r=1}^{k+1} u_r$ $= \sum_{r=1}^k u_r + u_{k+1}$ $= k(k+1)^2 + (k+1)(3k+4)$ $= (k+1)(k^2 + k + 3k + 4)$ $= (k+1)(k^2 + 4k + 4)$ $= (k+1)(k+2)^2$ $= RHS$ Since P_1 is true and P_k is true $\Rightarrow P_{k+1}$ is true. Therefore, by mathematical induction, P_n is true $\forall n \in \mathbb{Z}^+$.	[M1] [M1] [B1/2]
5	$\frac{dy}{dx} = \frac{x \frac{du}{dx} - u}{x^2}$ Substituting, $\frac{x \frac{du}{dx} - u}{x^2} = \frac{2u - 2\frac{u}{x} - 1}{x(x+2)}$ $x \frac{du}{dx} = u + \frac{2ux - 2u - x}{(x+2)}$ $x \frac{du}{dx} = \frac{3ux - x}{(x+2)}$ $\frac{du}{dx} = \frac{3u - 1}{(x+2)}$ $\int \frac{1}{3u-1} du = \int \frac{1}{x+2} dx$ $\frac{1}{3} \ln 3u-1 = \ln x+2 + \ln c$ $\ln 3xy-1 = \ln c(x+2) ^3$ $3xy-1 = k(x+2)^3$ $y = \frac{k(x+2)^3 + 1}{3x}$	[B1] [M1] [M1] [M1] [M1] [B1] [A1]

7	$\pi r^2 h + \frac{1}{2} \left(\frac{4}{3} \pi r^3 \right) = 500$ $\pi r^2 \left(h + \frac{2}{3} r \right) = 500$ $h = \frac{500}{\pi r^2} - \frac{2}{3} r$ $S = \pi r^2 + 2\pi r h + \frac{1}{2} (4\pi r^2)$ $= 3\pi r^2 + 2\pi r \left(\frac{500}{\pi r^2} - \frac{2}{3} r \right)$ $= \frac{5}{3} \pi r^2 + \frac{1000}{r}$ $\frac{dS}{dr} = \frac{10}{3} \pi r - \frac{1000}{r^2}$ $= \frac{10\pi r^3 - 3000}{3r^2}$ $\frac{dS}{dr} = 0 \Rightarrow 10\pi r^3 - 3000 = 0$ $\Rightarrow r = \sqrt[3]{\frac{300}{\pi}} \quad \text{or} \quad 4.5708 \quad (4 \text{ dp})$ $\frac{d^2 S}{dr^2} = \frac{10}{3} \pi + \frac{2000}{r^3} > 0 \quad \text{when} \quad r = \sqrt[3]{\frac{300}{\pi}}$ $\text{Minimum surface area} = \frac{5}{3} \pi \left(\frac{300}{\pi} \right)^{\frac{2}{3}} + 1000 \sqrt[3]{\frac{\pi}{300}}$ $= 328.1715 \quad (4 \text{ dp})$ Cost of box with minimum surface area $= (328.1715)(0.015) = 4.9226 = \\$4.92 \quad (2 \text{ dp})$	[M1] [M1] [M1] [M1] [M1] [M1] [M1] [M1] [M1] [A1]
8a	The statement is only true if p is real. (i) Using GC, $p = 5$. (ii) We have $z^4 - z^3 + 4z^2 + 3z + 5 = (z - (1 - 2i))(z - (1 + 2i))(z^2 + az + b)$, where a and b are real. $= (z^2 - 2z + 5)(z^2 + az + b)$ Comparing coefficients of similar terms, we have $a = b = 1$	[B1] [B1] [M1] [B1]

	For $z^2 + z + 1 = 0$, we have $z = -\frac{1}{2} \pm \frac{\sqrt{3}}{2}i$	[A1]
(b)	$\left \frac{a^3}{a^*} \right = \frac{ a ^3}{ a } = R^2$ $\arg(q) = \arg(a^3) - \arg(a^*)$ $= 3 \arg(a) + \arg(a)$ $= 4\alpha$ Thus, $q = R^2 [\cos(4\alpha) + i \sin(4\alpha)]$ $\Rightarrow q^{\frac{1}{6}} = R^{\frac{1}{3}} \left[\cos\left(\frac{2}{3}\alpha\right) + i \sin\left(\frac{2}{3}\alpha\right) \right]$ Given that $\cos\left(\frac{2}{3}\alpha\right) = 0$, $\frac{2}{3}\alpha = \frac{\pi}{2} \Rightarrow \alpha = \frac{3\pi}{4} \text{ or } 2.36 \text{ radians}$	[B1] [M1] [A1] [M1] - arg [M1] [A1]
9	(i) $\frac{dy}{dt} = \frac{-2t^2 + 2a}{(t^2 + a)^2}$, $\frac{dx}{dt} = 8t$ $\frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{dt}{dx} = \frac{a - t^2}{4t(t^2 + a)^2} = 0$ $t = \pm \sqrt{a}$ $x = 4a, y = \frac{\pm(2\sqrt{a})}{2a} = \pm \frac{\sqrt{a}}{a}$ Turning points: $(4a, \frac{\sqrt{a}}{a}), (4a, -\frac{\sqrt{a}}{a})$. (ii) $y = \frac{2t}{t^2 + a}$ $y^2 = \left(\frac{2t}{t^2 + a} \right)^2$	[M1/2] [M1/2] [M1/2] – dy/dx [M1/2] – “=0” [A1] [A1] [M1]

	$y^2 = \frac{x}{\left(\frac{x}{4} + a\right)^2}$ $y^2 = \frac{16x}{(x + 4a)^2} \text{ (shown)}$ (iii) $y^2 = \frac{16x}{(x + 4a)^2}$  (iv) Method 1: A translation of $4a$ unit in the direction of positive x-axis followed by a scaling of $1/3$ units along the x-axis. Method 2: Scaling of $1/3$ units along the x-axis followed by a translation of $4a/3$ unit in the direction of positive x-axis. (Accept use of the word “shift”)	[M1] Shape – B1 [–1/2 if sharp at origin] Turning pts – B1/2 each $(0, 0)$ – B1/2 “$y = 0$” – B1/2 Each transf – B1 [No ½ mark]
10a	GP : $a = 5, r = 2$	
(i)	$S_n = 5(2^n - 1) \leq 2500$ $\Rightarrow n \leq 8.97 \text{ (3 sf)}$ No. of complete bags to offer her friends = 8	[M1] - S_n [M1] - ineq [A1]
(ii)	No. of cookies left for sale = $2500 - 5(2^8 - 1) = 1225$ AP : $a = 4, d = 3$ $S_n = \frac{n}{2}(8 + 3(n - 1)) \leq 1225$ $\Rightarrow -29.4 \leq n \leq 27.8 \text{ (3 sf)}$ Last complete bag = 27^{th} bag $T_{27} = 4 + 26(3) = 82$	[M1] [M1] [B1] [M1]

	Cost of last complete bag = $82(0.5) = \$41$	[A1]
b(i)	$\ln T_r - \ln T_{r-1} = (2 + rb) \ln 2 - (2 + (r-1)b) \ln 2$ $= b \ln 2, \text{ a constant (shown)}$	[M1] [B1] – $b \ln 2$
(ii)	$S_n = \sum_{r=1}^n \ln T_r = \frac{n}{2} [2(2+b) \ln 2 + (n-1)b \ln 2]$ $= \frac{n \ln 2}{2} [(n+1)b + 4]$	[M1] [A1]
(iii)	$ S_{14} - S_{13} \leq 0.5 \Rightarrow \left (7 \ln 2)(15b + 4) - \frac{13 \ln 2}{2} (14b + 4) \right \leq 0.5$ $\Rightarrow -0.194 \leq b \leq -0.0913 \quad (3 \text{ sf})$ [No marks if modulus sign is left out]	[M1] [A1]
11i	Let $\mathbf{n}_1$ and $\mathbf{n}_2$ be the normals of p_1 and p_2 respectively.	
	$\mathbf{n}_1 = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} \times \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}$	[M1]
	$\mathbf{n}_2 = \begin{pmatrix} 2 \\ 3 \\ 0 \end{pmatrix} \times \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \\ -2 \\ -3 \end{pmatrix}$	[M1]
	$\begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} \times \begin{pmatrix} 3 \\ -2 \\ -3 \end{pmatrix} = \begin{pmatrix} 5 \\ 6 \\ 1 \end{pmatrix}$ Therefore l is parallel to $5\mathbf{i} + 6\mathbf{j} + \mathbf{k}$.	[M1]
(ii)	$\text{Acute angle between } p_1 \text{ and } p_2 = \cos^{-1} \frac{\left \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 3 \\ -2 \\ -3 \end{pmatrix} \right }{\sqrt{3}\sqrt{22}}$ $= 75.7^\circ$	[M1] [A1]
(iii)	$\text{Perpendicular distance} = \frac{\left \left[\begin{pmatrix} 4 \\ 2 \\ 2 \end{pmatrix} - \begin{pmatrix} 2 \\ 4 \\ 6 \end{pmatrix} \right] \cdot \begin{pmatrix} 3 \\ -2 \\ -3 \end{pmatrix} \right }{\sqrt{22}}$	[M1]

	$= \frac{\begin{vmatrix} 2 & 3 \\ -2 & -2 \\ -4 & -3 \end{vmatrix}}{\sqrt{22}}$ $= \frac{ -22 }{\sqrt{22}}$ $= \sqrt{22}$	[A1]
(iv)	The point (2, 4, 6) must lie on p_3 So $2a - 8 + 12 = b$ $b = 2a + 4$ Since the three planes meet at a single point, we must exclude the case where they meet along a line. If p_3 meets in the line l, the normal of p_3 is perpendicular to l. So $\begin{pmatrix} a \\ -2 \\ 2 \end{pmatrix} \cdot \begin{pmatrix} 5 \\ 6 \\ 1 \end{pmatrix} = 0$ $a = 2$ Therefore $b = 2a + 4$, where $a \neq 2$	[M1] [A1]
(v)	$p_1: \mathbf{r} \cdot \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} = 4$ and $p_3: \mathbf{r} \cdot \begin{pmatrix} 2 \\ -2 \\ 2 \end{pmatrix} = b$ Distance between p_1 and $p_2 = \frac{1}{\sqrt{3}}$ $\left \frac{4}{\sqrt{3}} - \frac{b}{2\sqrt{3}} \right = \frac{1}{\sqrt{3}}$ $\frac{4}{\sqrt{3}} - \frac{b}{2\sqrt{3}} = \frac{1}{\sqrt{3}}$ or $\frac{4}{\sqrt{3}} - \frac{b}{2\sqrt{3}} = \frac{-1}{\sqrt{3}}$ $b = 6$ or 10	[M1] [M1] [A1]